II. Short answer questions

1. What is the microscopic origin of pressure?
2. What is the microscopic origin of temperature?
3. Why moon has no atmosphere?
4. Write the expression for rms speed, average speed and most probable speed of a gas molecule.
5. What is the relation between the average kinetic energy and pressure?
6. Define the term degrees of freedom.
7. State the law of equipartition of energy.
8. Define mean free path and write down its expression.
9. Deduce Charles' law based on kinetic theory.
10. Deduce Boyle's law based on kinetic theory.
11. Deduce Avogadro's law based on kinetic theory.
12. List the factors affecting the mean free path.
13. What is the reason for Brownian motion?

III. Long answer questions

1. Write down the postulates of kinetic theory of gases.
2. Derive the expression of pressure exerted by the gas on the walls of the container.
3. Explain in detail the kinetic interpretation of temperature.
4. Describe the total degrees of freedom for monoatomic molecule, diatomic molecule and triatomic molecule.
5. Derive the ratio of two specific heat capacities of monoatomic, diatomic and triatomic molecules
6. Explain in detail the Maxwell Boltzmann distribution function.
7. Derive the expression for mean free path of the gas.
8. Describe the Brownian motion.

IV Numerical Problems

1. A fresh air is composed of nitrogen N_2(78%) and oxygen O_2(21%). Find the rms speed of N_2 and O_2 at 20°C.

   Ans: For N_2, $v_{rms} = 511\ m\ s^{-1}$
   For O_2, $v_{rms} = 478\ m\ s^{-1}$
2. If the rms speed of methane gas in the Jupiter's atmosphere is 471.8 m s^{-1}, show that the surface temperature of Jupiter is sub-zero.

   Ans: -130°C
3. Calculate the temperature at which the rms velocity of a gas triples its value at S.T.P. (standard temperature $T_1 = 273$ K)

   Ans: $T_2 = 2457$ K
4. A gas is at temperature 80°C and pressure $5 \times 10^{-10} N m^{-2}$. What is the number of molecules per m^3 if Boltzmann's constant is $1.38 \times 10^{-23} J K^{-1}$

   Ans: 1.02×10^{11}
5. If 10^{20} oxygen molecules per second strike 4 cm^2 of wall at an angle of 30° with the normal when moving at a speed of 2×10^3 m s^{-1}, find the pressure exerted on the wall. (mass of one oxygen atom = 2.67×10^{-26} kg)

   Ans: 46.2 N m^{-2}
6. During an adiabatic process, the pressure of a mixture of monatomic and diatomic gases is found to be

UNIT-9(XI-Physics_Vol-2).indd 185 26-03-2022 17:47:49